



Greenhouse Gas Emissions in Chicago: Trends and Analysis

BY: CIF SCIENTISTS

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Project Vision

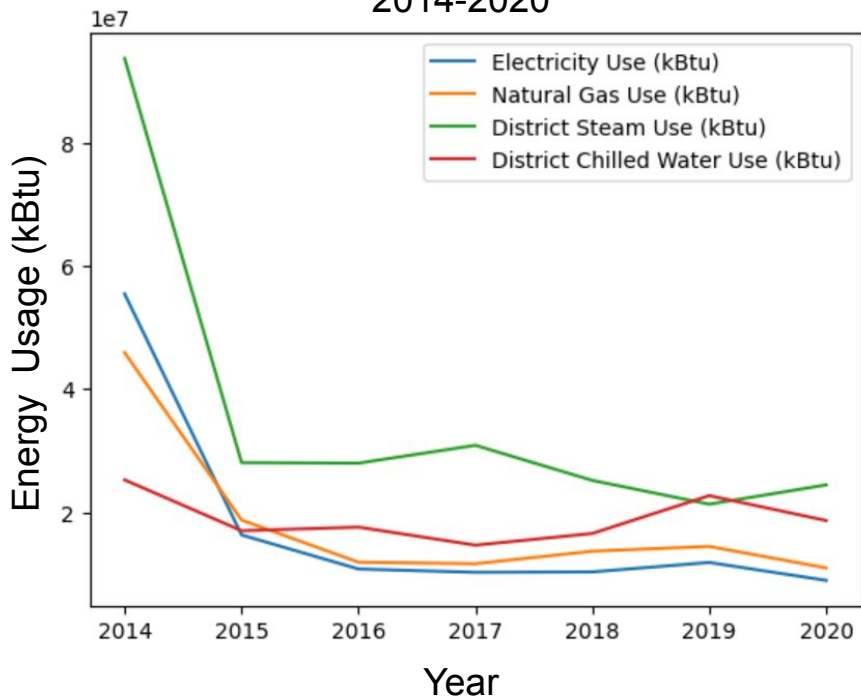


Use data science techniques to:

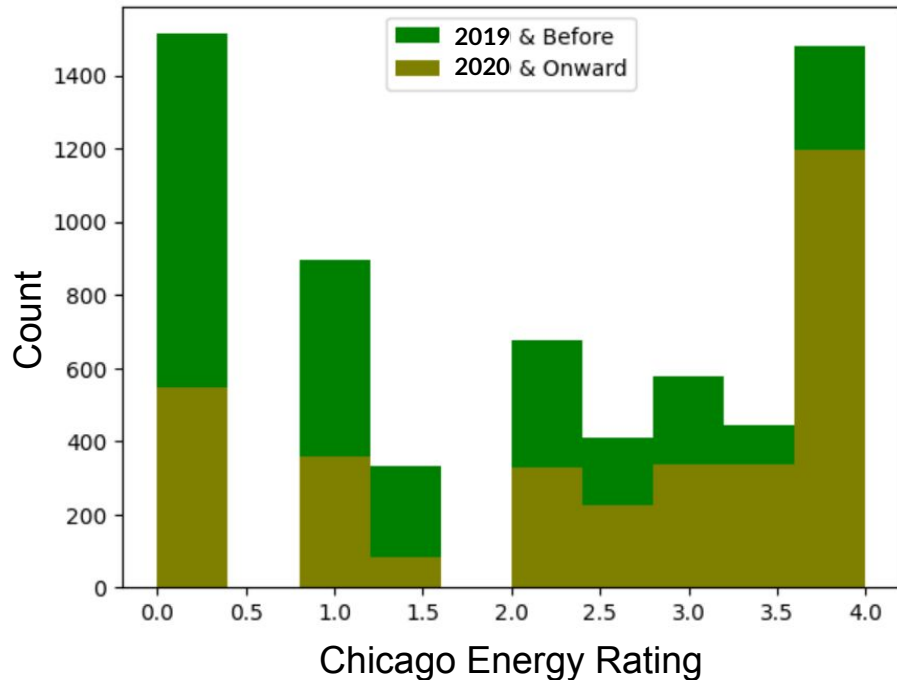
- Analyze and visualize the greenhouse gas emissions data in Chicago
- Identify the major sources of emissions
- Provide data from which recommendations for reducing emissions to help mitigate the effects of climate change can be created.

Descriptive analytics- How has energy use improved over time

Different Energy Usage from
2014-2020



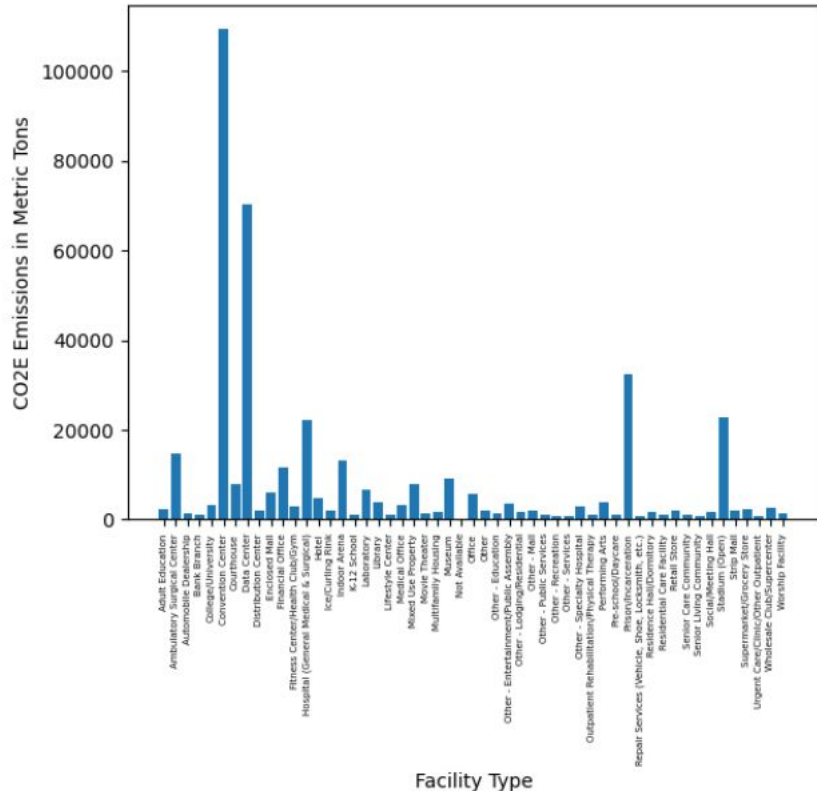
Comparing Chicago
Energy Rating



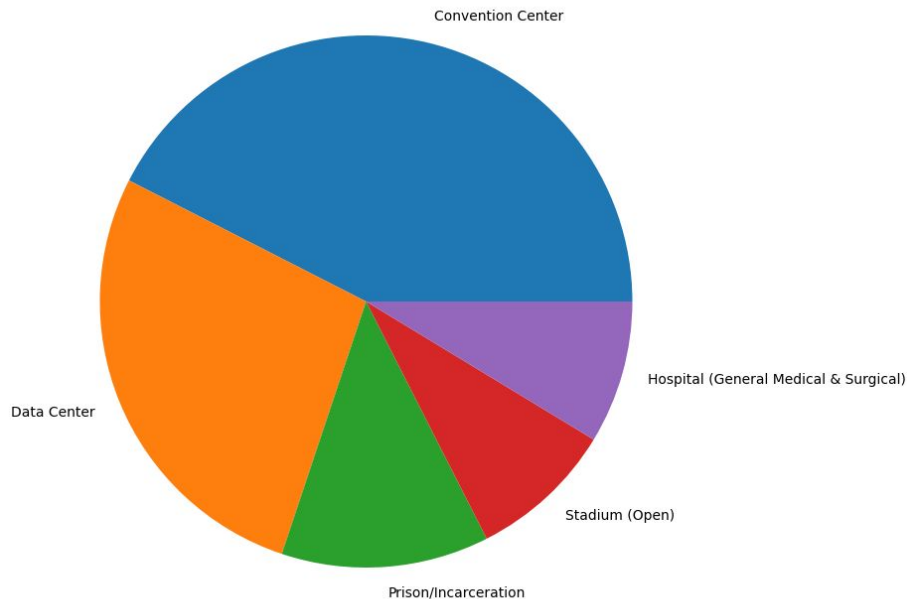
Chicago Building Energy Use Benchmarking
Ordinance of 2014

Descriptive analytics- What facilities types produce the highest emissions?

Average Emissions by Facility Type



Average Emissions (Top 5 Highest Polluting Facility Types)

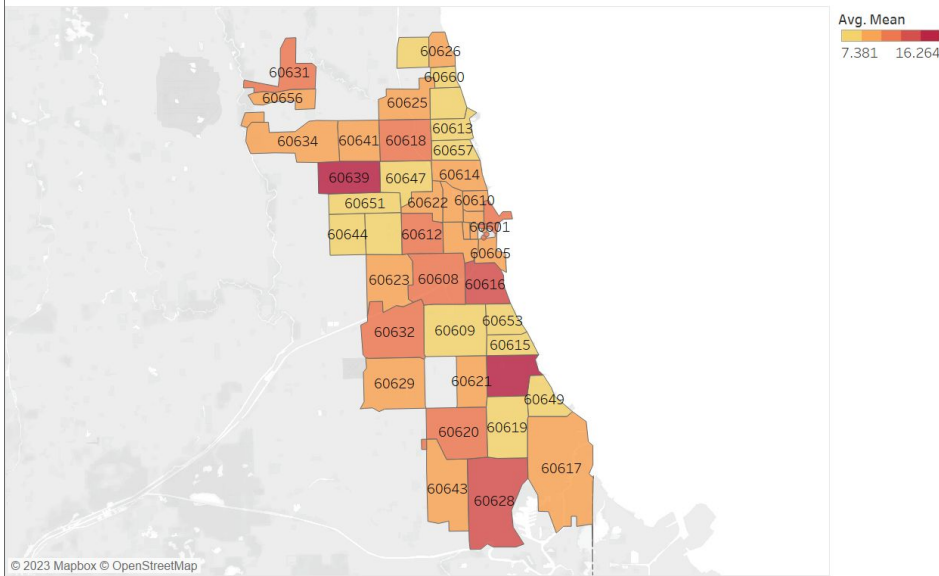


Convention centers, data centers, hospitals, stadiums, and prisons produce the highest emissions

Distance to Chicago Vs Greenhouse Gas Intensity

Result

Mean GHG Usage (sq. ft.) by Zip Code



© 2023 Mapbox © OpenStreetMap

Map based on Longitude (generated) and Latitude (generated). Color shows average of Mean. The marks are labeled by Zip Code.

Correlation Coefficient:
-0.2546

P value:
0.08072078407271438

Property Type's Contribution to GHG Intensity

Zip_Code	Primary_Property_Type	Count	Zip_Code	Primary_Property_Type	Count
60616	Multifamily Housing	362	60637	Multifamily Housing	134
60616	College/University	61	60637	Laboratory	121
60616	K-12 School	45	60637	K-12 School	63
60620	K-12 School	63	60639	K-12 School	58
60620	Multifamily Housing	27	60639	Retail Store	21
60620	Retail Store	13	60639	Wholesale Club/Supercenter	12
60620	Supermarket/Grocery Store	13			
60628	K-12 School	62			
60628	Multifamily Housing	25			
60628	Worship Facility	10			

Is there a significant difference between the Mean GHG emissions by building purpose?

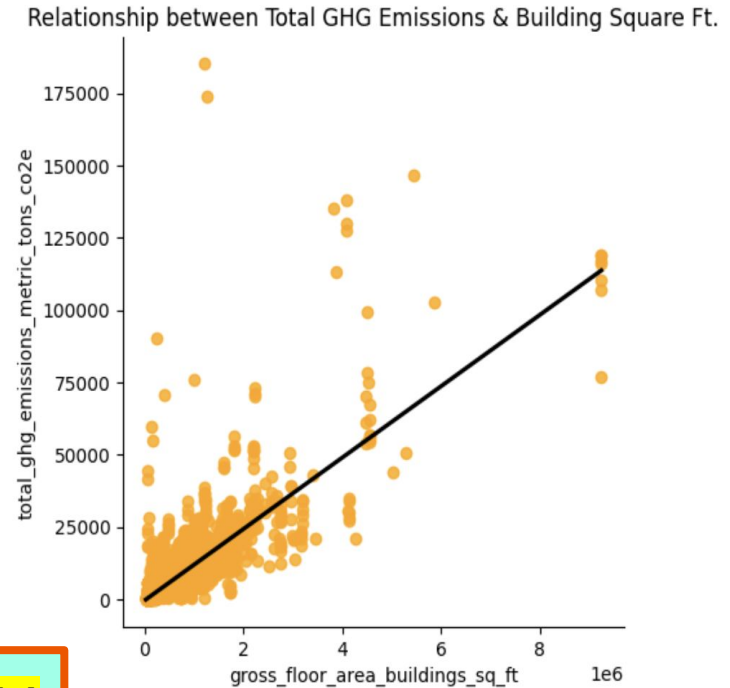
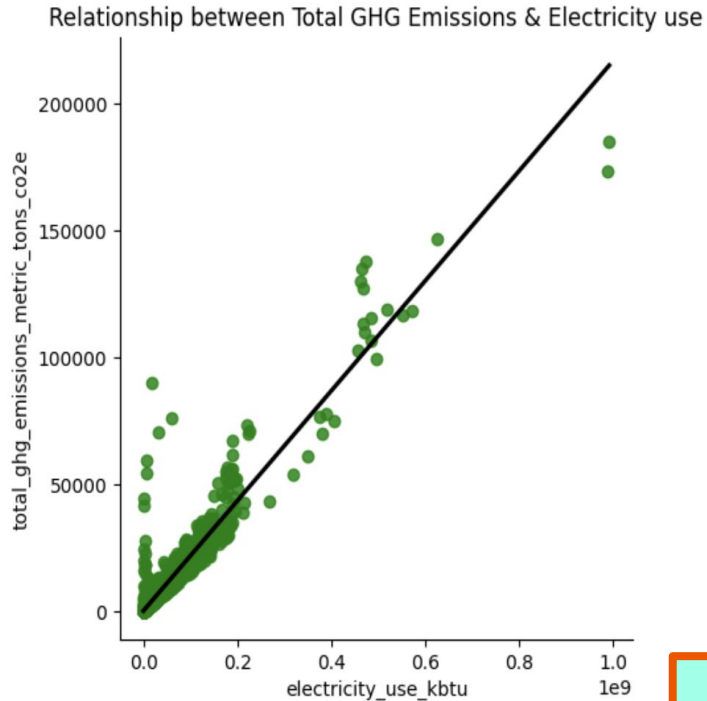

* = $p < 0.05$

	Vs. Commercial	Vs. Education	Vs. Entertainment	Vs. Others	Vs. Residential
Healthcare	0.758	*	0.871	0.976	0.302

Key Points

- **Initial thoughts:** Type of building might matter -> Healthcare buildings
- **Post-analysis conclusions:** healthcare related buildings emit significantly different GHG than educational buildings (educational buildings emit significantly more GHG emissions than Healthcare buildings)
- Educational buildings make up over 30% of the buildings we assessed, may be emitting significantly bigger carbon footprint in Chicago

What factors affect the amount of greenhouse gas emissions?



$$r^2 = .914$$

Can we predict the size of a building from energy use? - Classification

Features:

- Electricity Use
- Natural Gas Use

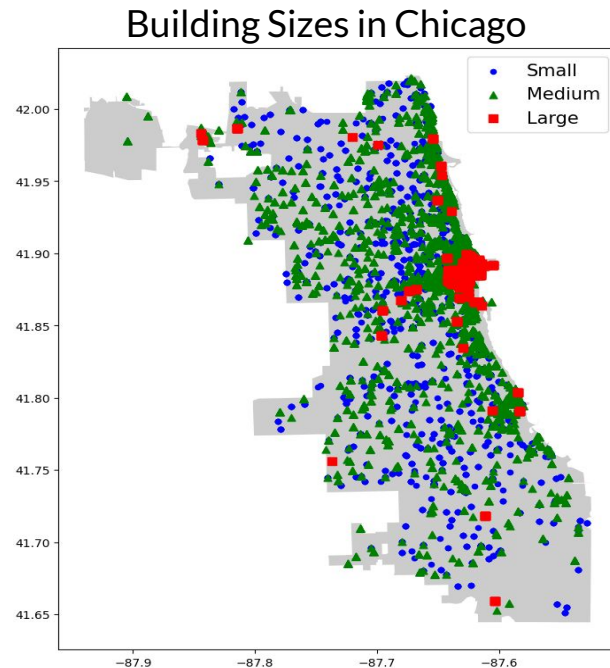
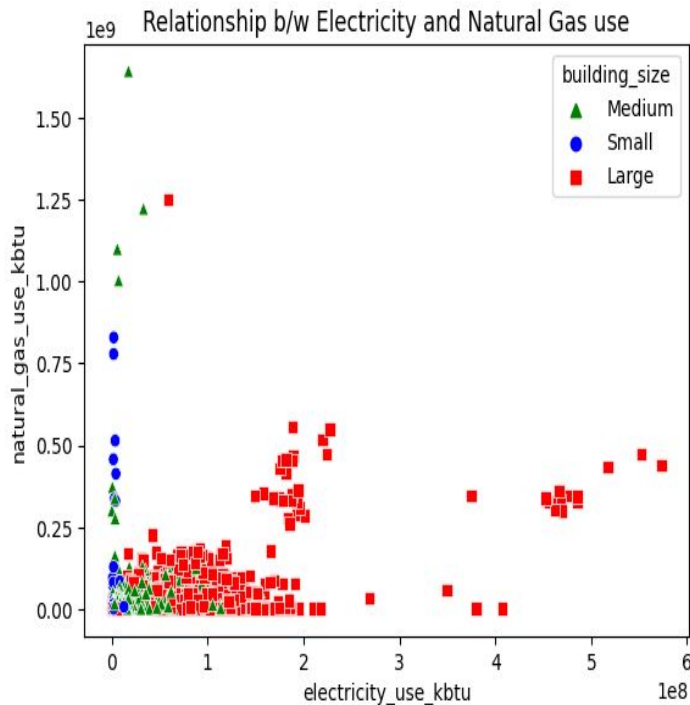
Label:

- Building Size

Random Forest Classification

Accuracy Score: 0.81

Cohen's Kappa: 0.64



Conclusion



The nature of cities contribute a significant amount of carbon emissions to the atmosphere. Chicago is one such city in America which has committed itself to reducing its carbon footprint.

Takeaways:

- Since the introduction of Chicago's Energy Benchmarking program, a significant reduction in energy use can be documented
- A strong relationship between the size of buildings in square feet and GHGs was uncovered.
- Chicago's energy program is effective at reducing GHGs.